Kienan Ma

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Education

University of Waterloo – *Bachelor of Computer Science, Honours Co-op* Sep. 2020 – Aug. 2025

Technical Skills

Languages: TypeScript, JavaScript, Kotlin, HTML/CSS, SQL, C++, C#, Python, Java
Technologies: React, Redux, Tailwind, Node, Express, Auth0, .NET, Jest, Cypress, MongoDB, PostgreSQL, SQLite, Next.js, Terraform, Vite, GitHub Actions, Docker, Git, Claude Code, Cursor
Certifications: AWS Certified AI Practitioner (2026)

Experience

Software Engineer Intern Jan. 2025 – Apr. 2025
BioRender *Toronto, ON*

- Architected and delivered a custom file management workspace using TypeScript and React for 4000+ active enterprise users, implementing role-based access controls, achieved 85% test coverage via Jest and Cypress
- Diagnosed and resolved critical account takeover vulnerability in authentication layer, shipped fix with full regression test coverage within 5 days, preventing potential unauthorized access for thousands of users
- Consolidated enterprise login routes into a unified authentication portal, simplifying user sign up and login flows; reduced authentication-related support tickets by 20% and cut user drop-off by ~10% during onboarding
- Leveraged Auth0 breach detection to flag 500+ public compromised credentials, implemented proactive user notifications, enforced mandatory password resets to secure affected accounts

Software Engineer Intern May 2024 – Aug. 2024
Scispot (YC21) *Kitchener, ON*

- Spearheaded comprehensive redesign of laboratory software including dashboards, file explorer, and spreadsheet editor using React, JavaScript, Tailwind, and Next.js, improving data visualization for 1000+ scientists
- Developed a library of 15+ reusable React UI components with comprehensive documentation, streamlining development process and standardizing design language, reducing front-end development time by ~20%

Software Engineer Intern Jan. 2023 – Apr. 2023
PINpoint Information Systems *Burlington, ON*

- Built a dynamic form generator using TypeScript, Angular, and JSONSchema for automated schema-driven UI generation and validation, eliminated 100% of manual form configuration across 20+ data entry modules, accelerating feature deployment from weeks to days
- Implemented and tested new features for manufacturing execution system software in .NET and C#, leveraged LINQ to simplify data querying and manipulation, reducing average query execution time by 25%

Frontend Developer Intern May 2022 – Aug. 2022
blueRover *Cambridge, ON*

- Modernized the frontend by migrating from Node 7 to Node 16, then refactored 20% of React class components to functional components with hooks, allowing adoption of modern libraries
- Optimized slow-running MySQL queries through index tuning, query restructuring for weekly internal and customer reports, reducing generation time by 30%, enabling faster business insights

Frontend Developer Intern Sep. 2021 – Dec. 2021
Metagenom Bio Life Science *Waterloo, ON*

- Built from scratch and deployed a web app using JavaScript, Node.js, and Postgres to assign and visualize lineage for 50+ plant virus strains, automating previously manual lineage assignments